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Ensemble averaging of symbolic numbers: Criterion mean judgments beat heuristics

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Abstract

Can observers compute an ensemble mean from *symbolic* numbers, or is averaging-like performance driven by heuristic shortcuts? We addressed this question using a criterion-based design that yields clear predictions for mean-based integration. In two experiments, participants viewed a 2×2 array of four Arabic digits and decided, while timed, whether the *arithmetic mean* of the set was smaller or larger than 5. In Experiment 1 ($N = 40$), accuracy and response time showed a robust ensemble-distance effect: decisions were slowest and least accurate when the set mean was closest to 5 and improved monotonically with mean distance. Item-level model comparisons further indicated that mean distance explained behavior better than prominent heuristic alternatives based on the most extreme digit or the majority of digits above versus below the criterion, and within-set dispersion produced an additional, independent impairment. Experiment 2 ($N = 40$) provided a stronger diagnostic test by constructing stimulus regimes that rendered heuristic cues uninformative (majority ties) or misleading (extreme-item conflict) while matching mean distance and dispersion across regimes. Across regimes, model comparisons again favored mean distance for both accuracy and response time. Critically, the distance effect persisted when majority information carried no diagnostic value and when the most extreme digit favored the incorrect response. These results support genuine mean-based integration of symbolic magnitudes, while showing that heuristic strategies have only a secondary role apparent only under increased uncertainty.

Open Practice Statement Data and stimuli can be found here https://osf.io/xnq92, and none of the experiments was preregistered.

Keywords Numerical cognition · Ensemble coding · Averaging · Distance effect · Decision-making

Introduction

A large body of work shows that observers can efficiently encode *summary statistics* of sets (Ariely, 2001; Chong & Treisman, 2003), including ensemble *averaging* over basic visual dimensions (e.g., size, orientation, luminance) (Dakin & Watt, 1997; Marchant, Simons, & de Fockert, 2013; Rajendran, Maule, Franklin, & Webster, 2021) and over higher-level attributes such as facial emotion and gender (Fitousi, 2025; Haberman & Whitney, 2007), as well as numerosity (Burr, & Ross, 2008; Halberda, Sires, & Feigenson, 2006). A natural extension is to ask whether ensemble averaging applies to *symbolic* inputs, such as Arabic numerals, whose magnitudes are not physically given but learned through abstract mappings (Algom, Dekel, & Pansky, 1996; Fitousi, 2014; Pansky, & Algom, 2002).

Several studies already suggest that observers can estimate the mean of symbolic numbers (Brezis, Bronfman, Jacoby, Lavidor, & Usher, 2016; Brezis, Bronfman, & Usher, 2015; Brezis, Bronfman, & Usher, 2018; Levin, 1974; Rosenbaum, Glickman, & Usher, 2021; Tsetsos, Chater, & Usher, 2012). However, the main question is whether the averaging-like patterns observed reflect genuine integration of numerical magnitude or, alternatively, capture the usage of simpler heuristics (Gigerenzer & Gaissmaier, 2011; Tversky & Kahneman, 1974). In particular, earlier studies have used multi-digit numbers (Brezis et al., 2015; Tsetsos et al., 2012), which are subject to place-value decomposition (Dehaene, Dupoux, & Mehler, 1990; Fitousi, & Algom, 2020; Nuerk, Weger, & Willmes, 2001) and related attentional operations (Fitousi & Algom, 2020). Participants can use strategies that take advantage of the symbolic structure

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inherent in those numbers and respond accurately without fully integrating the items in the set. They can do so by such heuristics as anchoring to landmarks, selective sampling, and extreme-majority-based rules. Moreover, even within traditional averaging-based accounts, performance can be affected by processes that are not specific to the mean (Myczek & Simons, 2008). For example, Tsetsos et al. (2012) reported strong recency weighting and sensitivity to within-set variance. The upshot is that averaging-like patterns in those studies can reflect a mixture of true integration, memory dynamics, and dispersion-based influences, rather than a pure computation of the set mean.

The present study addresses this problem by deploying experimental designs that are grounded in numerical-cognition theory (Dehaene, 2011; Moyer & Landauer, 1967; Verguts & Fias, 2004). These designs allow us to test not only mean-based integration and averaging, but also to detect whether participants use heuristics and strategy-based shortcuts. First, we restrict stimuli to *single-digit* Arabic numerals, which allows us to eliminate place-value parsing (Dehaene et al., 1990; Hinrichs, Yurko, & Hu, 1981) and decomposition-based operations (Fitousi & Algom, 2020; Nuerk et al., 2001). Second, we present the set *simultaneously* in a static display (visible until response) rather than as a sequential stream. This allows us to remove temporal order effects (e.g., recency weighting) and reduce the reliance on working-memory dynamics – factors that may be a main source of confound in stream-based averaging (Brezis et al., 2015; Tsetsos et al., 2012). Third, we exploit the *numerical distance effect* (Dehaene, 2011; Dehaene et al., 1990; Fitousi & Algom, 2020; Moyer & Landauer, 1967) – a robustly documented phenomenon in comparative judgments of numerals, capturing a monotonic improvement in performance as a function of numerical distance. The distance effect has been often interpreted as reflecting the transformation of symbolic numerals to analog internal quantities. We take advantage of this classic effect to derive quantitative predictions for ensemble averaging. Specifically, if participants form an analog representation of the set mean, performance should improve with mean-distance evidence (e.g., $\Delta = |\bar{x} - 5|$ in a set-to-criterion task).

Our approach therefore offers a window into the processes that support averaging-like performance with symbolic ensemble displays. Specifically, it tests whether symbolic numerals are transformed into analog magnitude signals that can be integrated across items, or whether participants instead rely on heuristics afforded by the symbolic structure of the display (Campbell, & Austin, 2002; Dehaene, Piazza, Pinel, & Cohen, 2005; Fitousi, 2010; Fitousi, & Algom, 2006; Fitousi, Shaki, & Algom, 2009; Gigerenzer, & Gaissmaier, 2011; Kadosh, & Walsh, 2009; Lemaire, & Siegler, 1995; Reyna, Nelson, Han, & Dieckmann, 2009; Tversky, & Kahneman, 1974; Verguts, & Fias, 2004). The experimental designs used here allow these two modes of processing to be placed in direct competition. Our goal is to distinguish genuine mean-based integration from plausible shortcut strategies, and thereby to link symbolic averaging both to theories of ensemble perception (Rosenbaum, de Gardelle, & Usher, 2021) and to accounts of symbol-to-magnitude mapping in numerical cognition (Dehaene, 2003; Dehaene, Bossini, & Giraux, 1993; Fitousi, 2014; Izard, & Dehaene, 2008; Nieder, Freedman, & Miller, 2002; Zorzi, Priftis, & Umiltà, 2002).

Averaging symbolic numbers

Existing demonstrations of symbolic-number averaging come from three related, but conceptually distinct, classes of paradigms. The first uses *single-stream mean estimation*, in which a series of digits is presented one-after-the-other with a constant temporal separation (Brezis et al., 2016, 2015, 2018; Corbett, Oriet, & Rensink, 2006; Rosenbaum, Glickman, Fleming, & Usher, 2022; Rosenbaum, Glickman, & Usher, 2021; Tsetsos et al., 2012). For example, Brezis et al. (2015) presented sequences of 4, 8, or 16 two-digit numbers in rapid serial order and asked participants to report the sequence average. Their goal was to distinguish between putatively analytic and intuitive modes of numerical averaging as a function of set size and presentation rate. This work is highly relevant because it shows that observers can derive approximate average magnitude from temporally extended symbolic input, even under demanding presentation conditions. However, it is less diagnostic with respect to the present question. This is mainly because, in this work, participants are asked to produce an estimate. Consequently, good performance can emerge due to the deployment of heuristics rather than due to a full mean-based integration. For example, participants can sample only the decade digits and ignore the units (Dehaene et al., 1990), use partial sampling by relating only to the smallest and largest numbers, selectively weight a subset of items (Myczek & Simons, 2008), anchor on salient values (Tversky & Kahneman, 1974), or approximate a running sum. Thus, these studies speak primarily to the *capacity* and *mode* of symbolic averaging, but are less suited to answering whether behavior is best explained by the set mean rather than by potential shortcut strategies.

In a second class of paradigms, observers compare *two sets* and decide which has the larger average (Corbett et al., 2006; Levin, 1974; Rosenbaum et al., 2022; Tsetsos et al., 2012). This literature spans highly simplified displays (Corbett et al., 2006) as well as more complex rapid streams (Tsetsos et al., 2012). For example, Corbett et al. (2006) showed that participants could discriminate between two simultaneously presented arrays composed only of 2 s and 5 s, reliably selecting the array with the higher average. This finding is crucial because it shows that participants can rapidly extract

numerical information from symbolic arrays. However, the stimuli in this task are restricted to two possible numerals, and therefore participants can perform well in this task without actually computing the average of the set. For example, participants can use majority voting rather than compute the arithmetic mean across all elements.

Note that in these paradigms participants can compare the sums of the two sets instead of their arithmetic averages. The problem even worsens when researchers present long sequences of numerals. In this case, participants can take advantage of additional heuristics, such as selective subsampling, over weighting of salient extremes, or relying on diagnostic samples. These tasks are important because they show that participants can summarize sets of symbolic numbers. However, they are less suitable for answering the question raised in the present study of whether participants can average the numerical magnitude of the numerals in a set.

An illustrative example comes from the study by Tsetsos et al. (2012). These researchers administrated several clever manipulations to rule out simple heuristics (e.g., choosing the stream with the single maximum). This led them to argue against a fixed-capacity subsampling account. However, their analyses revealed strong temporal weighting (recency) and sensitivity to variance. These patterns are consistent with leaky accumulation and salience-based overweighting, rather than with an unweighted arithmetic mean. This entails that performance in such sequential symbolic averaging tasks can reflect a mixture of integration, memory dynamics (e.g., recency), and dispersion/salience effects. The upshot is that novel designs are needed that can disentangle mean-based integration from strategy-based alternatives.

A third class of studies presents participants with a *single* array of numbers, and asks participants whether its average is smaller or larger than a fixed reference (Vandormael, Herce Castañón, Balaguer, Li, & Summerfield, 2017). These tasks bear close affinity with the criterion-based mean judgment deployed in the present study. These tasks define a fixed boundary, and thus invite analysis of how performance changes as a function of distance from the boundary. However, these tasks too were not designed to optimally adjudicate between true averaging and heuristics. For example, Vandormael et al. (2017) presented *eight* two-digit numbers under free eye movements. They used this task to study *sampling policy* and the weighting of inlying versus outlying items during perceptual choice. Their results therefore speak primarily to which items receive greater weight once observers sample from a large display, rather than to whether performance in symbolic mean comparison is best explained by genuine mean integration as opposed to simpler shortcut rules. Indeed, because such displays contain many items and allow selective inspection, they naturally permit strategies based on partial sampling, selective attention, or disproportionate reliance on salient values.

Related work using *single simultaneous displays* (Rosenbaum, de Gardelle, & Usher, 2021) has used *mean estimation* tasks. Observers reported the average of a digit set on a continuous response scale. These tasks are highly informative about precision and capacity limits, but they are similarly restricted with respect to mechanism. For example, good estimates can arise from subsampling, anchoring, mid-range rules, or selective reliance on extreme values (Gigerenzer & Gaissmaier, 2011; Tversky & Kahneman, 1974). These heuristics can mimic the true mean without requiring full integration of the set.

The present study

The goal of the present study is to address the question of whether participants can average symbolic numerals without relying on heuristics or strategic shortcuts. We used a comparison task with a fixed standard. The task and attendant stimuli offer strong diagnostic tools, as they allow us to pit genuine averaging against heuristic performance in an explicit and controlled fashion. On each trial, participants viewed a static set of Arabic digits and indicated, while timed, whether the *arithmetic mean* of the set was smaller or larger than a fixed reference value (5). The display remained visible until response. This task provides two main advantages. First, it takes advantage of the well-established distance effect (Moyer & Landauer, 1967), and gives a clear quantitative indicator of whether participants computed an integrated representation of the set mean. Specifically, performance should get improved as the distance of the overall set's mean from the standard of 5 ($\Delta = |\bar{x} - 5|$) gets larger. This is consistent with the well-established idea in numerical cognition that numerals are transformed onto their analog representations, and the comparison operation is applied to those noisy representations (Dehaene et al., 1990). We expect that averaging of symbolic numbers is governed by a similar process, by which numerals are transformed into noisy representations, integrated, weighted, and then compared to the standard. Second, because the criterion is held fixed, we can design stimulus sets in which Δ is systematically dissociated from prominent shortcut cues (e.g., the most extreme digit or the majority of digits above vs. below 5). This will enable us a direct test of mean-based integration versus heuristic substitutes.

We harnessed a four-item display with a 2×2 array of single-digit Arabic numerals. This choice complies with two methodological demands. First, the task must require integra-

tion across multiple symbolic values. Second, the stimulus space must remain tightly controllable to afford the dissociation of mean-based evidence from potential heuristics. One needs to manipulate within-set dispersion and heuristic cues independently of mean-based evidence. Note that larger arrays (e.g., 3×3 or 4×4) make it very hard to construct balanced sets. Moreover, pilot work revealed that these displays yielded poorer performance, likely due to heavier demands on memory. The 2×2 format therefore provides an optimal display. It is large enough to require multi-item integration, but small enough to support precise experimental control and interpretable tests of averaging versus heuristics.

The goal of Experiment 1 was to test our novel paradigm, to assess the presence of a distance effect in averaging, and to confirm that participants indeed track the mean of the set. The goal of Experiment 2 was to provide stronger support for this hypothesis. In this experiment, we engineered trials classes in which prominent heuristic shortcuts either became uninformative or predicted the wrong response. In one type of trials, the numerals were evenly split around the fixed criterion of 5. This eliminated any informative majority cue. In another type of trials, the most extreme digit fell on the opposite side of the criterion from the set mean. So, if participants base their decision on the extreme value, they would choose the incorrect response. Thus, the experiments conducted here address a stronger question than prior work. They ask not only whether participants can summarize symbolic sets, but whether participants track the actual set mean when potential computational heuristics and shortcuts are neutralized or put in direct conflict with it.

Overview and hypotheses

We applied the same comparison judgment task with a fixed standard of 5 in both Experiment 1 and Experiment 2. Participants indicated whether the average of the set was smaller or larger than 5. Our central index for averaging-like behavior was the distance between the set mean and the reference value $\Delta = |\bar{x} - 5|$.

Experiment 1 established the basic psychometric and chronometric signatures of performance under naturalistic variation in set composition. Its primary goal was to assess whether trial-by-trial behavior is better explained by mean-distance evidence, as predicted by mean-based integration, or by simpler properties of the display that correlate with the mean.

Experiment 2 provided a more stringent test by constructing stimulus regimes that approximate a factorial manipulation. Mean distance was restricted to a small set of values, within-set dispersion was systematically varied, and trials were assigned to diagnostic classes that either neutralized or opposed specific shortcuts. This enabled a direct test of whether mean tracking persists precisely when non-averaging strategies should fail.

Across both experiments, we tested three linked hypotheses. First, the *distance hypothesis*: accuracy should increase and response times should decrease monotonically with Δ. Second, the *dispersion hypothesis*: holding Δ constant, greater within-set variability should impair performance (higher error rates and slower responses), consistent with noisier integration. Third, the *averaging-versus-heuristic hypothesis*: trialwise behavior should be better accounted for by mean-distance evidence than by prominent shortcut cues based on especially salient or more frequent items; critically, in Experiment 2, the influence of Δ should remain evident even in the diagnostic classes designed to undermine these shortcuts. If, instead, observers rely primarily on non-averaging shortcuts, then the mean-distance effect should be substantially reduced or absent in the diagnostic classes where those shortcuts are uninformative or misleading, and performance should be better predicted by the corresponding heuristic cues than by Δ.

Experiment 1

Method

Participants

Forty adults ($N = 40$, 18 females and 22 males, mean age = 29.12, sd = 12.18) from the local community of Ariel and the Ariel participant pool took part. All reported normal or corrected-to-normal vision and provided informed consent. The experiments received the approval of the Ethic Committee of Ariel University.

Stimuli

Each display comprised four integers drawn (with replacement) from 1, 2, 3, 4, 6, 7, 8, and 9, i.e., excluding the digit 5. Digits appeared in black near the four corners of a mid-gray square (see Fig. 1). Let $x = (x_1, x_2, x_3, x_4)$ denote the four digits, $\bar{x} = \frac{1}{4}\sum_{i=1}^{4} x_i$ the set mean, and $\Delta = |\bar{x} - 5|$ the absolute distance from the decision criterion. The stimulus set sampled 14 mean-of-set levels, $\bar{x} \in \{1.5, 2.0, 2.5, \ldots, 4.5, 5.5, \ldots, 8.0, 8.5\}$, that is, integer half-steps from 1.5 to 8.5 excluding 5.0, producing symmetric distance values $\Delta \in \{0.5, 1.0, \ldots, 3.5\}$. The position of the four digits within the 2×2 array was randomly permuted for each item to avoid positional regularities. The complete set of stimuli contained 69 unique displays, each presented multiple times across participants. Stimuli were pre-generated offline and saved as image files. The full set of all 69 stimuli (ensembles) and their composition can be downloaded from https://osf.io/xnq92.

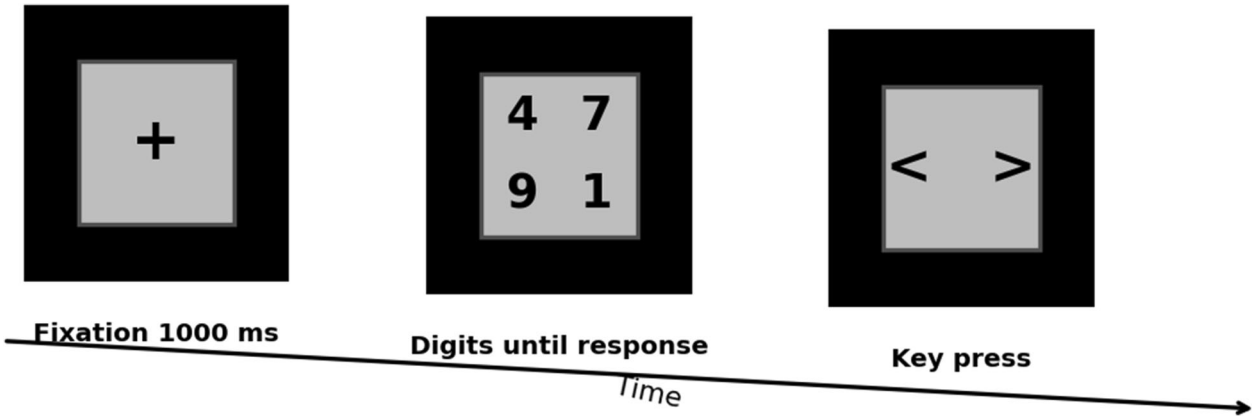


Fig. 1 Trial structure. Each trial began with a central fixation cross (1000 ms), followed by the stimulus display. The display consisted of four black digits arranged in a 2×2 grid on a gray square within a black panel. Participants judged whether the mean of the four digits was smaller or greater than 5

Apparatus and Procedure

We conducted the experiments in our laboratory on laptop computers. On each trial, a fixation point was presented for 1000 ms, and then a 2×2 array of four Arabic digits was presented. Stimuli were presented until a response. Participants indicated by keypress whether the *arithmetic mean* of the four numbers was smaller or greater than 5 (see Fig. 1). They pressed a left key if the average of the set of smaller than 5, and a right key if it was larger than 5. Participants were encouraged to estimate the mean of the set and avoid computing the mean in an algorithmic fashion. Both speed and accuracy were highlighted. Trials were randomized, and no feedback was given. Importantly, we kept the numerals visible until response in order to align the task with standard symbolic numerical comparison paradigms (Moyer & Landauer, 1967). In those paradigms, stimuli remain available and difficulty is driven primarily by the comparison signal rather than by short-term memory constraints. Our relevant theoretical question is not whether observers *could* compute exactly, but which decision variable best predicts accuracy and response time under the canonical conditions of symbolic comparison (Dehaene et al., 1990).

Design

The design was within-participants. The primary manipulated factor was the mean-of-set (fourteen levels as above), or equivalently the absolute Δ distance from the criterion. The side of the mean relative to the criterion ($\bar{x} < 5$ vs. $\bar{x} > 5$) was balanced across trials. Because the four digits were drawn with replacement from $\{1, 2, 3, 4, 6, 7, 8, 9\}$, the within-set dispersion $\mathrm{sd}_4 = \mathrm{sd}(x_1, x_2, x_3, x_4)$ varied naturally across items but was not independently manipulated in this experiment.

Participants completed four blocks, each consisting of 69 trials. On each trial, one of the 69 possible stimuli was presented in random order without replacement. In total, each participant completed 276 trials.

Results

The total error rate amounted to 4.8%. Trials with anticipations or extreme latencies were trimmed using bounds of 150 ms $\leq$ RT $\leq$ 3800 ms. This procedure removed 470 of 11,360 trials (4.14%). Trimmed trials were excluded from RT analyses and retained as errors for accuracy summaries.

Distance effect Mean RT decreased linearly with the absolute distance from the criterion, $\Delta = |m - 5|$. A linear regression on bin means yielded an excellent fit, $R^2 = .932$, with a strongly negative slope, $b_\Delta = -193.51$ ms per unit Δ (95% CI [−226.44, −160.59]), $t(12) = -12.80$, $p < 0.005$. The intercept at $\Delta = 0$ ($m = 5$) was 1375.34 ms (95% CI [1301.72, 1448.97]); residual SE = 56.55 ms.

Error rates decreased sharply with Δ. A binomial logistic regression on aggregated counts showed a strongly negative distance slope, $b_\Delta = -1.110$ (SE = 0.061), $z = -18.15$, $p < 0.005$, implying that each unit increase in Δ multiplies the odds of error by $\exp(b_\Delta) = 0.330$ (95% CI [0.292, 0.372]). At $\Delta = 0$ ($m = 5$), the implied baseline error rate was $p_0 = 0.218$ (95% CI [0.191, 0.248]). Model fit indices were: null deviance = 497.44 on 13 df, residual deviance = 61.35 on 12 df, AIC = 134.35.

Gaussian shape of RTs and error rate

To augment the distance effect analyses and to capture the expected inverted-U shape pattern of performance (Dehaene et al., 1990; Fitousi & Algom, 2020), we also fit a symmet-

ric Gaussian to binned outcomes $y(m)$ (mean RT or error proportion):

$$y(m) = a + b \exp\Big(-\frac{(m-\mu)^2}{2\sigma^2}\Big), \quad (1)$$

where a is the baseline (tail) level, $b > 0$ the peak height above baseline, μ the center (predicted near 5), and σ the width. The results (see Fig. 2(a)-(b)) showed canonical inverted-U centered on the criterion: $a = 767.75$ ms (95% CI [733.55, 801.96]), $b = 615.83$ ms (95% CI [571.60, 660.05]), $\mu = 4.994$ (SE = 0.0365; 95% CI [4.922, 5.065], not different from 5), and $\sigma = 1.202$ (SE = 0.064; 95% CI [1.077, 1.326]). The implied peak at the center is $a+b \approx 1383.58$ ms. Applying Eq. 1 to error proportions revealed a low tail $a = 0.016$ (95% CI [0.010, 0.020]) and a central peak $a+b = 0.203$ ($b = 0.187$; 95% CI [0.170, 0.204]). The fitted center was $\mu = 4.86$ (SE = 0.024; 95% CI [4.814, 4.907], significantly below 5), with width $\sigma = 0.678$ (SE = 0.031; 95% CI [0.618, 0.738]). The Gaussian pattern supports the chronometric effect, indicating maximal difficulty near the criterion and improved performance with increasing distance.

These numerical distance patterns are very similar to those obtained in comparative judgment tasks with two-digit numbers, where participants are presented with a two-digit number and indicate whether it is larger or smaller than a predefined standard (e.g., 55) (Dehaene et al., 1990; Fitousi & Algom, 2020; Hinrichs et al., 1981). In the case of two-digit numbers, these profiles are often taken as strong evidence that the decade and unit digits were integrated into a holistic

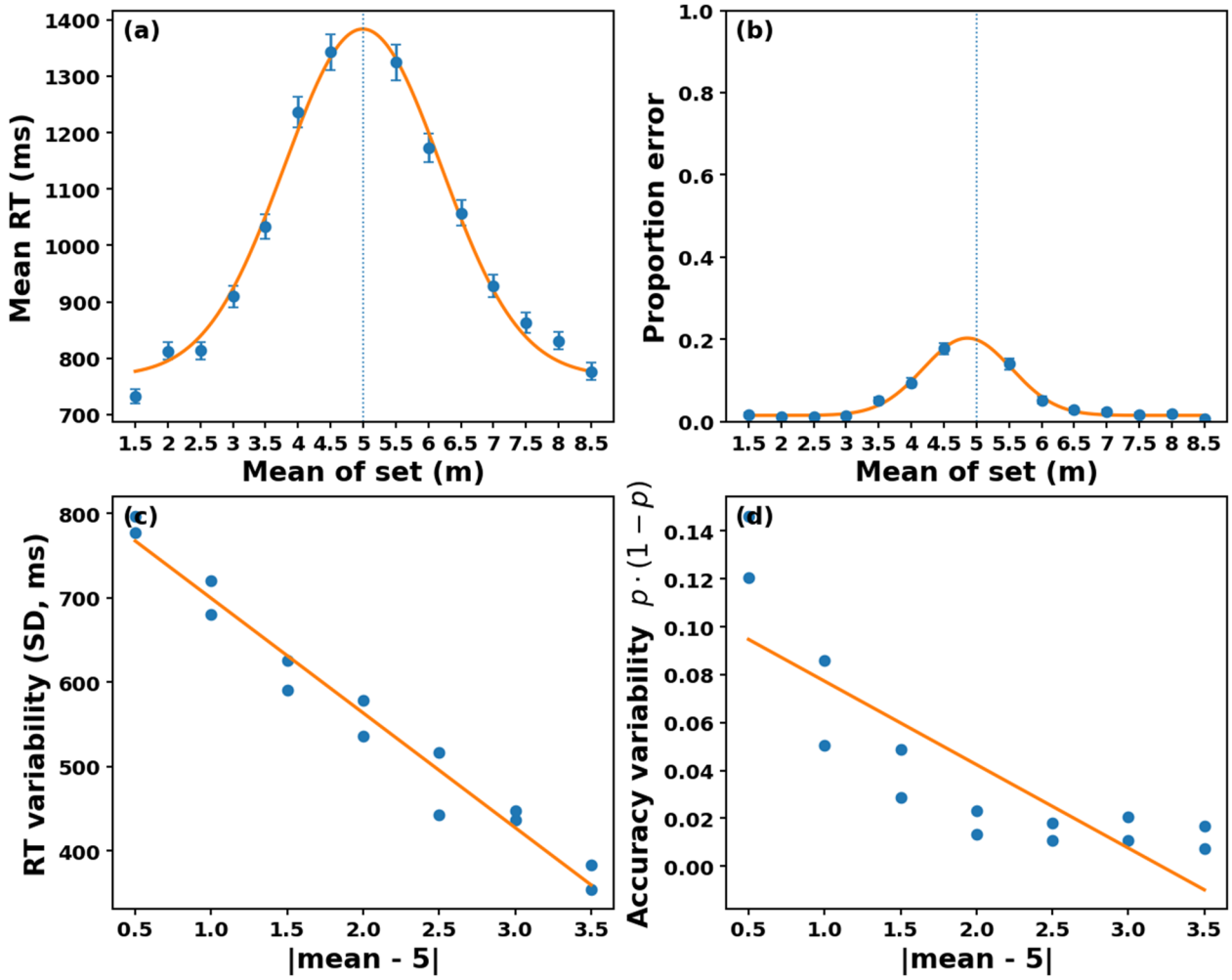


Fig. 2 Summary of performance and variability. **(a)** Mean RT (ms) as a function of the display mean m; points are bin means at the 14 mean-of-set levels (1.5, 2.0, . . . , 8.5) with ±1 SE *bars*, the *solid curve* is a Gaussian fit, and the *vertical dotted line* marks the criterion $m = 5$. **(b)** Proportion of errors as a function of m with the same conventions. **(c)** RT variability (within-bin SD) as a function of distance from the criterion, $\Delta = |m - 5|$; the *line* shows a fitted linear trend. **(d)** Accuracy variability (Bernoulli variance $p(1-p)$) as a function of Δ; the *line* shows a fitted linear trend. Panels (a) and (b) display a bell-shaped difficulty profile centered on the criterion, while panels (c) and (d) show decreasing variability with increasing distance from the criterion

analog representation. The patterns observed with our single-digit ensembles can be interpreted according to a similar logic. The four digits were mapped onto their analog quantity, integrated and compared with the reference, hence the continuous and monotonic distance effect.

Variability analyses

We next examined whether response variability also changed as a function of distance from the reference of 5, which is the category bound (see Fig. 2(c)-(d)). For response times, the standard deviation of RT decreased reliably with distance, as shown by a strong negative linear effect, $b = -135.78$, $SE = 7.20$, $t(12) = -18.87$, $p < .001$, $R^2 = .967$. The intercept was 835.01 ($SE = 16.09$), indicating maximal RT variability near the boundary and progressively lower variability for stimuli farther away.

We conducted a comparable analysis of response variability in accuracy using the Bernoulli variance term, $p(1-p)$, weighted by the number of observations in each bin. This analysis likewise revealed a significant negative effect of distance, $b = -0.035$, $SE = 0.007$, $t(12) = -5.05$, $p < .001$, $R^2 = .680$. The intercept was 0.112 ($SE = 0.016$). Thus, both RT variability and choice variability were highest near the reference of 5 and declined as stimuli became more distant from it.

Individual-level consistency of the distance effect

To examine whether the distance effect reflected a broadly shared pattern or was driven by a small subset of participants, we repeated the distance analyses at the individual-participant level. For each participant, we estimated the relation between mean distance and error rate, and separately the relation between mean distance and log-transformed RT on correct trials. The pattern was highly consistent across observers. For accuracy, all 40 participants showed the predicted negative relation between Δ and error rate, indicating fewer errors as the set mean moved farther from the criterion. This consistency was reliable by an exact sign test, $p < .001$, and the median participant-level slope was -1.34. Because some participants made very few errors, ordinary logistic regressions produced separation-related instability for a small number of individual fits; we therefore emphasize the directional consistency and median slope rather than the mean of the individual logistic slopes.

The RT results showed the same participant-level consistency. All 40 participants showed the predicted negative relation between Δ and log RT, indicating faster responses as the set mean moved farther from the criterion. This pattern was again reliable by an exact sign test, $p < .001$. The median participant-level RT slope was -0.17, and the mean slope differed reliably from zero, $t(39) = -13.72$, $p < .001$, 95% CI $[-0.204, -0.152]$. Thus, the basic distance effect was not confined to a subset of participants, but was present across the full sample for both accuracy and response time.

Averaging vs. heuristic alternatives

Predictors and rationale We compared three item-level predictors of performance that distinguish an averaging account from simpler heuristic alternatives. First, the absolute distance of the *mean* from the decision criterion 5,

$$\Delta = |\overline{x} - 5|\,, \tag{2}$$

captures a mean-based averaging account. Second, the *maximum* absolute deviation from 5 among the four digits,

$$M_{\max} = \max_i |x_i - 5|\,, \tag{3}$$

captures an extreme-value rule in which decisions are driven by the most deviant item in the set. Third, the *majority* count above versus below 5, operationalized as the absolute sum of signs,

$$M_{\text{maj}} = \left|\sum_{i=1}^{4} \text{sgn}(x_i - 5)\right|, \tag{4}$$

captures a vote-count rule in which responses depend on how many items fall on each side of the criterion. By design, the digit 5 never appeared and no display had $\overline{x} = 5$, so $\text{sgn}(x_i - 5) \in \{-1, +1\}$ on every trial, and majority ties arise only in 2–vs.–2 displays.

We also included within-set dispersion, sd_4, defined as the sample standard deviation of the four digits.

$$\text{sd}_4 = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2} \qquad \text{with } n = 4. \tag{5}$$

This covariate controls for spread at a fixed mean: two arrays can share the same mean (e.g., $m = 4.5$) yet differ substantially in dispersion. For example, $\{4, 4, 4, 6\}$ has $\text{sd}_4 = 1.00$, whereas $\{2, 4, 6, 6\}$ has $\text{sd}_4 \approx 1.92$. By using sd_4 we could separate the effect of the set mean from the effect of how widely the digits are distributed around that mean. Crucially, the dispersion variable allows us to test whether performance is fully determined by the mean-based decision variable, or whether it is also sensitive to the internal spread of the set. We hypothesized that if performance depends only on the final mean estimate relative to the criterion, then once Δ is held constant, within-set dispersion should contribute little or nothing. In contrast, if more dispersed sets make the mean less precise or harder to extract from the individual digits, then greater within-set spread should independently predict lower accuracy and longer RTs.

Model selection To compare averaging against heuristic alternatives, we fit three otherwise parallel mixed-effects models for each dependent measure. In each model, performance was predicted by one focal decision variable – mean distance, max distance, or majority vote – together with sd_4, and a random intercept for participant. The critical question was which focal predictor best accounted for performance beyond within-set dispersion.

For RTs, we fit linear mixed models to correct trials. Again, the mean-distance model provided the best fit (AIC = 157,828.9), outperforming the max model (ΔAIC = 43.2) and the majority model (ΔAIC = 109.4). For errors, we fit binomial mixed models with random intercepts for participants. The mean-distance model fit best (AIC = 3538.0), decisively outperforming the max model (ΔAIC = 32.8) and the majority model (ΔAIC = 38.3). Thus, across model-comparison criteria, performance was best captured by the distance between the set *mean* and the decision boundary.

Fixed-effect estimates for the winning model We next report the coefficients of the best-fitting (mean-distance) model.

In the RT model,

$$\mathrm{RT} = \beta_0 + \beta_\Delta\,\Delta + \beta_{\mathrm{sd}_4}\,\mathrm{sd}_4 + u_{\mathrm{sub}} + \varepsilon, \tag{6}$$

responses were faster farther from the criterion, $\beta_\Delta = -142.65 \pm 6.59$ ms, $t = -21.66$, and slower for more dispersed sets, $\beta_{\mathrm{sd}_4} = 129.99 \pm 9.37$ ms, $t = 13.88$. Thus, even after controlling for within-set dispersion, performance still tracked the distance between the set mean and the criterion. At the same time, the reliable effect of sd_4 shows that behavior was not determined by mean distance alone, but was also modulated by the internal spread of the set.

In the error model,

$$\mathrm{logit}\,P(\mathrm{error}) = \beta_0 + \beta_\Delta\,\Delta + \beta_{\mathrm{sd}_4}\,\mathrm{sd}_4 + u_{\mathrm{sub}}, \tag{7}$$

the distance slope was strongly negative, $\beta_\Delta = -1.011 \pm 0.076$, $z = -13.25$, $p < .001$. This corresponds to an odds multiplier of $\exp(\beta_\Delta) \approx 0.364$, indicating that each one-unit increase in Δ reduced the odds of error by approximately 64%. Within-set dispersion also impaired accuracy, $\beta_{\mathrm{sd}_4} = 0.161 \pm 0.073$, $z = 2.21$, $p = .027$.

Conflict diagnostics No displays produced a conflict between the mean's sign and the majority's sign (0 majority-conflict items), so that test was not estimable. For *max-conflict* displays (the extreme digit's sign opposed the mean; $n = 295$), reduced models were rank-deficient, indicating insufficient variation to estimate slopes within that subset. For *majority ties* (2 vs. 2; $n = 417$), Δ was constant ($\Delta = 0.5$ for all such items), so a distance slope was mathematically unidentifiable.

Discussion

For both speed and accuracy, the mean-based account appeared to fit the data better than the alternative heuristic accounts of majority items (above versus below criterion) or most extreme item. Moreover, within-set dispersion turned out to induce an independent cost on performance. Sets with higher dispersion tended to produce slower and less accurate responses. Taken together, these findings provide strong evidence against the possibility that participants deploy simple max-based or vote-count heuristics. They support the notion that participants are sensitive to the arithmetic mean value of the set and its variability, in a manner consistent with symbolic ensemble averaging (Brezis et al., 2016, 2015, 2018; Levin, 1974; Rosenbaum, Glickman, & Usher, 2021; Tsetsos et al., 2012).

However, at this stage, Experiment 1 should be interpreted as strong but not yet fully decisive evidence for averaging. The main result documented is a set-mean distance effect, namely, performance improved as the distance between the ensemble arithmetic mean and the fixed criterion increased. This is exactly what one would expect if participants indeed extract a summary representation of the set and compare it to the fixed standard. However, by itself, this pattern does not rule out the possibility that participants make use of simpler heuristics. Note that the numerals in the displays were not sampled in a way that made the relevant cues for heuristic performance independent of the set mean. For example, the mean's distance from the criterion, $\Delta = |\bar{x} - 5|$, tends to covary with other potentially useful cues, such as the side favored by the majority of items and the distance of the most extreme item. As a result, a heuristic process can partially mimic the same monotonic trend captured by the distance effect.

The variable of within-set dispersion revealed an important aspect of the experiment. Even after controlling for mean distance, more dispersed sets were associated with poorer performance. This means performance was governed not only by mean distance from the criterion, but also by the set's variance. This supports the idea that participants extract the set mean from the individual numerals, and that this process becomes more difficult as the set becomes more heterogeneous. However, Experiment 1 cannot by itself inform us exactly how this cost arises, because in this experiment we included the dispersion variable as a covariate and did not manipulate it as a primary design factor.

A further limitation is that the geometry of the stimulus space restricted the most diagnostic comparisons. There were no displays in which the mean and majority cues directly

conflicted, and majority ties were concentrated at a single value of Δ, making the critical within-subset slopes mathematically unidentifiable. Likewise, the available max-conflict items provided insufficient variation for stable slope estimation. Thus, although the mean distance effect obtained in Experiment 1 strongly favors a mean-based account, the properties of its stimuli space preclude a clear dissociation between averaging and heuristic alternatives.

A second experiment is needed that addresses these issues. In this experiment, we constructed displays that deliberately decorrelated the competing cues, so that the set mean, the majority cue, the extreme item, and within-set dispersion were more strongly dissociated. Only under such conditions can one test whether performance continues to track the set mean when the simpler heuristics are placed in direct competition with it. In this sense, Experiment 1 establishes the basic pattern of symbolic averaging, whereas the following experiment is needed to provide a sharper test of whether that pattern reflects genuine averaging rather than residual heuristic confounding.

Experiment 2

Experiment 2 was designed to provide a stronger test of symbolic averaging by addressing the possibility of heuristic confounds that remained in Experiment 1. Specifically, the stimulus set was constructed to vary mean distance and within-set dispersion more independently, while also placing the mean-based cue in direct relation to competing heuristic cues. To this end, displays were sampled from regimes in which the majority cue either agreed with the mean, was uninformative (majority ties), or in which the most extreme item opposed the mean (max-conflict). In all cases, the two sides of the criterion balanced overall.

This design more clearly tests whether performance is genuinely governed by the set mean. Under a mean-based averaging account, responses should continue to track the distance between the set mean and the criterion, Δ, even after controlling for within-set dispersion, sd_4. Crucially, this distance effect should remain evident in conditions where simpler heuristics lose their diagnostic value. For example, when the majority cue is neutral, or when the most extreme item points in the opposite direction from the mean. By contrast, if participants rely primarily on max-based or vote-count strategies, performance should shift toward those alternative cues precisely in such conflict conditions.

Experiment 2 will allow us to evaluate these alternatives by comparing several item-dependent models in which the focal predictor is, in turn, mean distance, the maximum absolute deviation from the criterion, or the majority count, while holding constant the remaining covariates. In this sense, Experiment 2 lets us go beyond the correlational evidence of Experiment 1. We can now ask a more diagnostic question: when competing cues are disentangled and, at times, forced into opposition, does performance still follow the set mean? If so, this would provide substantially stronger evidence that observers are performing genuine ensemble averaging of symbolic numbers rather than relying on simpler heuristics that merely covary with the mean under less constrained sampling.

Method

Participants

A new sample of adult volunteers (N = 40, 12 males 28 females, mean age = 21.65, sd = 2.56) took part for course credit. All reported normal or corrected-to-normal vision and provided informed consent in accordance with the Ariel University Ethics Committee.

Stimuli

Stimuli were four-digit sets sampled from the digits (1, 2, 3, 4, 6, 7, 8 ,9) (i.e., the reference digit 5 never appeared), so the mean distance from the decision boundary was never zero. For each set $X = \{x_1, x_2, x_3, x_4\}$, we computed the set mean, its distance from 5, and within-set dispersion (sd_4) as defined earlier (see Eq. 2; sd_4 was defined previously in Eq. 5).

To diagnose and rule out heuristic alternatives, stimulus construction was also guided by the *direction* of the majority cue and by the sign of the most extreme item. The majority direction was defined as the sign of the same sum used earlier to compute the majority-count predictor:

$$M = \mathrm{sgn}\Big(\sum_{i=1}^{4} \mathrm{sgn}(x_i - 5)\Big), \tag{8}$$

so $M \in \{-1, 0, +1\}$, with $M = 0$ indicating a 2–2 split around 5. The extreme-item sign was defined as

$$j^* = \arg\max_j |x_j - \bar{x}|, \qquad E = \mathrm{sgn}(x_{j^*} - 5), \tag{9}$$

where $E \in \{-1, +1\}$ indicates whether the most deviant digit lies below or above 5.

Design factors and stimulus selection From the full integer stimulus space, we selected a balanced subset of unique four-digit sets to populate three mutually exclusive *diagnostic regimes*, while also balancing mean-distance from the criterion (Δ), dispersion bin (low, med, high), and mean side ($\bar{x} < 5$ vs. $\bar{x} > 5$). Our goal was to construct a stimulus set that would be maximally informative for distinguishing a

mean-based averaging account from simpler heuristic alternatives.

The three diagnostic regimes were defined as follows:

$$\text{Regime} \in \begin{cases} majority_tie & \text{if } M = 0, \\ majority_agree & \text{if } M \neq 0 \text{ and the majority supports } \operatorname{sgn}(\bar{x} - 5), \\ max_conflict & \text{if the set contains a salient outlier digit opposing } \operatorname{sgn}(\bar{x} - 5). \end{cases} \tag{10}$$

Thus, majority_tie trials rendered the majority cue uninformative, majority_agree trials preserved agreement between the majority cue and the mean, and max_conflict trials created cases in which an extreme-item strategy would favor the response opposite to that implied by the set mean.

Within each populated cond_label × sd_label × Δ cell, the number of sets with $\bar{x} < 5$ was matched exactly to the number with $\bar{x} > 5$. The final stimulus list comprised 248 unique sets, distributed across regimes and factor levels as shown in Table 1. In every nonempty cell, counts were therefore perfectly balanced across the two sides of the criterion.

The variable of within-set dispersion was sampled at three design bins (low, med, high). In the final stimulus set, the observed sample standard deviations ranged from 0.577 to 2.517 in the low bin, from 1.291 to 3.000 in the medium bin, and from 1.732 to 4.123 in the high bin. The observed ranges partially overlapped because these were design bins rather than strictly nonoverlapping intervals on the continuous sd_4 scale. This reflects the combinatorial constraints of the stimulus space. Not all combinations of regime, dispersion level, and mean-distance are realizable, and some feasible combinations admit only a restricted range of dispersion values. Consequently, the final set was designed not to include every possible cell, but to provide the richest feasible and most balanced diagnostic sample for testing our competing hypotheses.

Table 1 Experiment 2: stimulus counts by diagnostic regime, dispersion bin (sd_label), and mean-distance bin $\Delta = |\bar{x} - 5|$. Counts pool across $\bar{x} < 5$ and $\bar{x} > 5$; within each nonempty cell, the split is equal across sides (i.e., $n/2$ per side). *Note.* Empty cells indicate condition × dispersion × distance combinations not included in the final stimulus set under the present sampling and balancing constraints, rather than missing observations

Regime	sd_4 bin	Δ = 0.5	Δ = 1.5	Δ = 2.5	Total
majority_agree	low	16	16	16	48
	med	0	16	16	32
	high	12	16	16	44
majority_tie	low	16	0	0	16
	med	16	12	0	28
	high	16	0	0	16
max_conflict	low	16	0	0	16
	med	16	0	0	16
	high	16	16	0	32
Total		**124**	**76**	**48**	**248**

The full set of 248 stimuli (ensembles) and their composition can be found here https://osf.io/xnq92.

Apparatus and procedure

The task was identical to Experiment 1. Participants sat in the lab in front of a laptop computer. On each trial, participants viewed a 2 × 2 array of four digits and indicated by keypress whether the *arithmetic mean* of the four numbers was smaller (left-hand key) or greater (right-hand key) than 5. Displays remained until response. Trials were randomized uniquely for each participant.

Design

The design crossed four within-participant factors. Mean distance from the criterion had three levels, $\Delta \in \{0.5, 1.5, 2.5\}$, presented symmetrically below and above 5. Within each Δ level, within-set dispersion sd_4 was binned by sample tertiles into low, medium, and high dispersion. The diagnostic regime had three levels (majority-agree, majority-tie, max-conflict), and side was coded as below vs. above the criterion ($\operatorname{sgn}(\bar{x}-5) \in \{-1, +1\}$). The resulting 3×3×3×2 structure was sampled approximately evenly (subject to availability in the integer stimulus space). The 248 stimuli were presented twice to each participant, once in each of two separate blocks, yielding a total of 496 trials per participant.

Results

Because Experiment 2 included diagnostically constructed trials that were more demanding than those in Experiment 1, including majority-tie and extreme-conflict trials, responses were generally slower. We therefore used a wider trimming window of 200 ms < RT < 5000 ms to avoid excluding valid responses on difficult trials. This cutoff excluded 6.6% of trials. For the RT analyses, incorrect trials were additionally excluded, removing 16.5% of the trials that remained after RT trimming.

Figures 3 and 4 present mean RTs and error rates, respectively, as a function of distance from the fixed standard and level of dispersion. Table 2 gives performance across

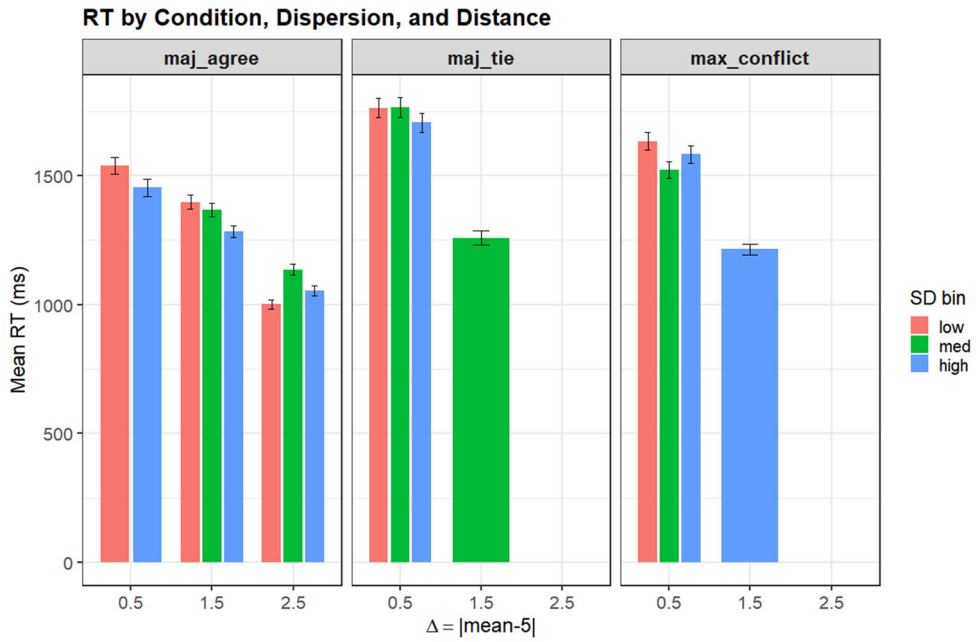


Fig. 3 Mean correct-trial response time (ms) in Experiment 2 as a function of mean-distance from the decision boundary (Δ) and within-set dispersion. *Bars* show mean RT for each Condition × SD × Distance cell, plotted separately for the three diagnostic regimes (majority_agree, majority_tie, and max_conflict). *Error bars* denote ±1 standard error

the observed experimental cells. Some cells are necessarily absent because they are not realizable within the stimulus space; however, the observed cells are sufficient to discriminate among the competing hypotheses. Across cells, accuracy increased and RT decreased as the set mean moved farther from 5. At the smallest distance from the criterion ($\Delta = 0.5$), mean accuracy ranged from .721 to .805 across the available cells, and mean correct RTs were relatively slow (approx-

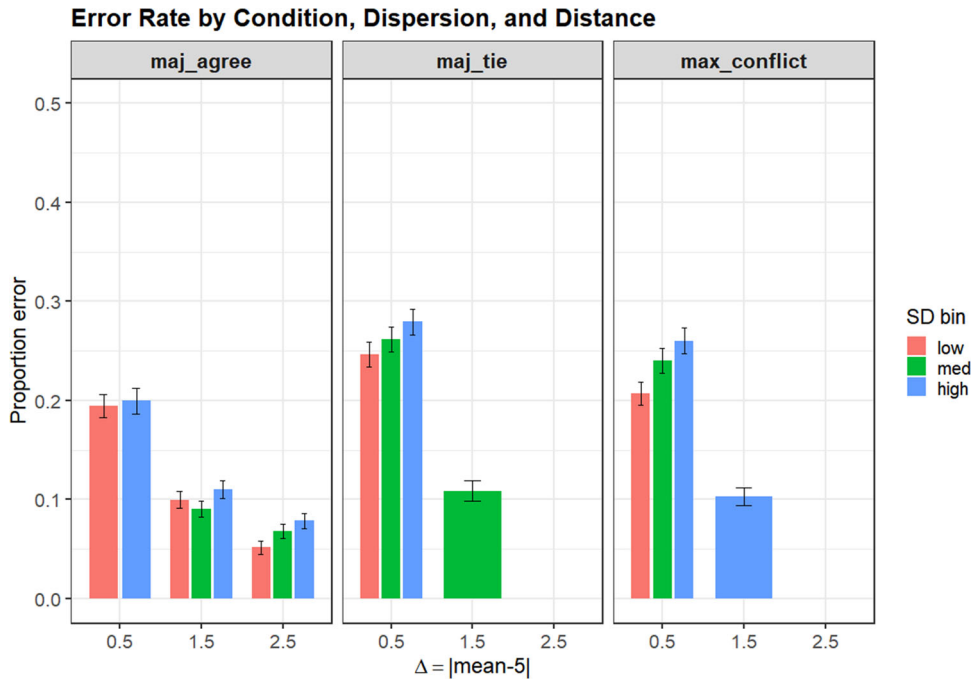


Fig. 4 Mean error rate in Experiment 2 as a function of mean-distance from the decision boundary (Δ) and within-set dispersion. *Bars* show the proportion of errors in each Condition × SD × Distance cell, plotted separately for the three diagnostic regimes (majority_agree, majority_tie, and max_conflict). *Error bars* denote ±1 standard error

Table 2 Accuracy and RT (ms) by Condition × SD × Distance (observed cells only)

Condition	SD	Distance	n	Accuracy	RT (ms)
maj_agree	low	0.5	1202	.805	1539
maj_agree	low	1.5	1215	.900	1396
maj_agree	low	2.5	1222	.948	1001
maj_agree	med	1.5	1218	.910	1368
maj_agree	med	2.5	1224	.932	1135
maj_agree	high	0.5	902	.800	1454
maj_agree	high	1.5	1219	.890	1283
maj_agree	high	2.5	1224	.922	1053
maj_tie	low	0.5	1165	.754	1762
maj_tie	med	0.5	1173	.738	1764
maj_tie	med	1.5	901	.891	1258
maj_tie	high	0.5	1181	.721	1706
max_conflict	low	0.5	1168	.793	1633
max_conflict	med	0.5	1206	.760	1523
max_conflict	high	0.5	1171	.740	1582
max_conflict	high	1.5	1215	.897	1213

Note. Condition codes index agreement structure within a trial: majority_agree = the majority of items support the same category as the aggregate/mean signal; majority_tie = equal support for the two categories (maximal ambiguity); max_conflict = maximal disagreement between the aggregate/mean signal and an extreme item (conflict trials). **SD** is the within-trial dispersion level (low, med, high). *Distance* is the absolute distance of the aggregate/mean from the decision boundary (5)

imately 1450–1760 ms). At larger distances, accuracy was high (often $\geq$ .89) and RTs were faster, typically around 1000–1280 ms in cells where $\Delta = 1.5$ or $\Delta = 2.5$ was present. Majority-tie trials at $\Delta = 0.5$ were associated with particularly slow RTs (approximately 1710–1760 ms) and comparatively increased error rates.

Distance effect Replicating Experiment 1, response times showed a strong monotonic distance effect when aggregating correct-trial RTs by mean-of-set. A linear regression of mean RT on distance from 5 yielded a large negative slope, $B = -282.90$ ms per distance unit, SE $= 35.49$, $t(4) = -7.97$, $p = .0013$, $R^2 = .941$ (95% CI for B: $[-381.44, -184.36]$ ms). Thus, responses were substantially faster as the mean moved farther from the decision boundary at 5. Errors likewise decreased sharply with distance. A binomial logistic regression on aggregated counts $\big($errors, correct$\big)$ as a function of distance produced $B = -0.831$, SE $= 0.032$, $z = -25.73$, $p < .001$. On the odds-ratio scale, each unit increase in distance reduced error odds by a factor of $\exp(B) = 0.44$ (approximately a 56% reduction; 95% CI $\approx [0.41, 0.47]$).

Gaussian shape of RTs and error rate Figure 5a, b show the descriptive relation between performance and mean-of-set, together with Gaussian fits summarizing the binned patterns. For RTs, the Gaussian fit to binned mean RTs was well behaved, yielding parameter estimates of $a = 1015.89$, $b = 694.54$, $\mu = 4.802$, and $\sigma = 1.100$. This fit captures the characteristic peaked RT profile centered near the decision criterion. For errors, the Gaussian fit to the raw error proportions was also very good, with parameter estimates of $a = 0.062$, $b = 0.238$, $\mu = 4.608$, and $\sigma = 0.729$. This function describes the elevated error rates near the criterion and their decline as the mean moved farther from the boundary. Notably, these Gaussian fits are used descriptively to summarize the shape of the binned functions; the main inferential conclusions continue to rest on the distance analyses and the mixed-effects model comparisons reported above.

Behavioral variability across distance At the descriptive level, behavioral variability also decreased with distance from the criterion (see Fig. 5(c)–(d). Across binned mean-of-set values, RT variability (within-bin SD) declined strongly with distance, $B = -173.91$, SE $= 12.39$, $t(4) = -14.04$, $p < .001$, $R^2 = .980$. Accuracy variability, indexed by the Bernoulli variance $p(1-p)$, also decreased with distance, $B = -0.06233$, SE $= 0.02001$, $t(4) = -3.11$, $p = .036$, $R^2 = .708$ (weighted by bin counts). These analyses are descriptive and parallel the pattern observed in Experiment 1.

Within-set dispersion as a function of distance Because within-set dispersion (sd_4) can covary with mean-distance (Δ) in discrete integer stimulus spaces, we examined this relation directly in the selected stimulus set. Within-set dispersion decreased reliably as distance from the criterion increased: stimulus-level sd_4 declined with Δ, $B = -0.5447$, SE $= 0.0626$, $t(246) = -8.70$, $p < .001$, $R^2 = .235$. Thus, sets nearer the decision boundary were, on average, also more internally dispersed. This dependence is a property of the admissible stimulus space rather than a behavioral result, and it motivates the inclusion of sd_4 as a control factor in all primary analyses.

Planned mixed-effects model comparison: averaging vs. heuristics

To adjudicate averaging (Δ) against heuristic alternatives, we fit planned trial-level mixed-effects models predicting (i) errors (logit link) and (ii) log RTs on correct trials, each with random intercepts for participant and controlling within-set dispersion. The primary averaging models were:

$$\text{logit}\, P(\text{error}_{ij} = 1) = \beta_0 + \beta_\Delta\, \Delta_{ij} + \boldsymbol{\beta}_{\text{sd}}^{\top}\mathbf{SD}_{ij} + u_{0j}, \quad (11a)$$

$$\log(\text{RT}_{ij}) = \gamma_0 + \gamma_\Delta\, \Delta_{ij} + \boldsymbol{\gamma}_{\text{sd}}^{\top}\mathbf{SD}_{ij} + v_{0j} + \varepsilon_{ij}, \quad (11b)$$

where $\Delta_{ij} = |\text{mean}_{ij} - 5|$, $\mathbf{SD}_{ij}$ codes sd_label (e.g., two contrasts relative to the reference level), and u_{0j}, v_{0j} are participant random intercepts.

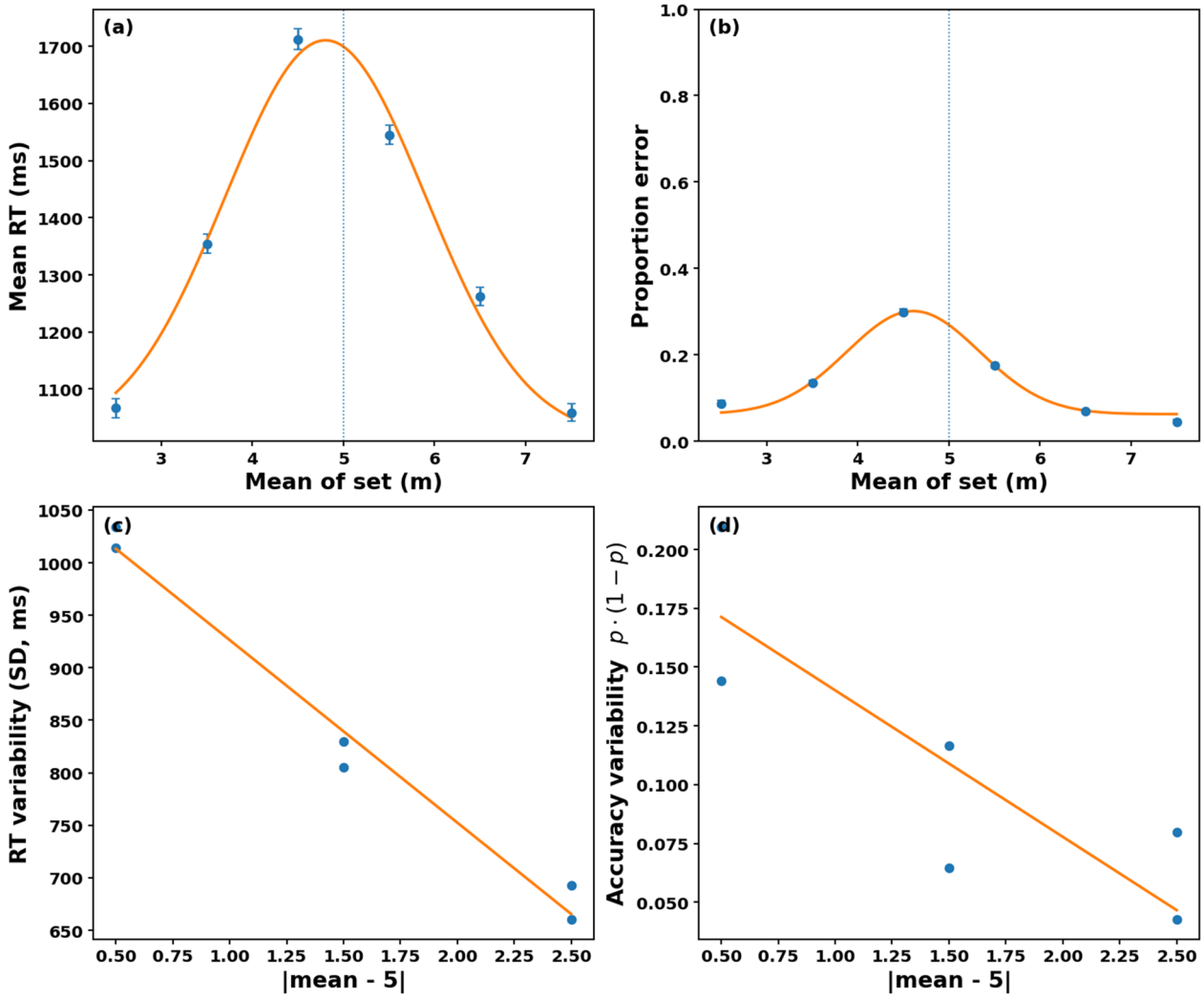


Fig. 5 Distance effects in Experiment 2: Reaction time, accuracy, and response variability as a function of the mean distance from the reference value. (**a**) Mean reaction time (RT) plotted against the mean of the number set (binned). *Error bars* represent standard errors. (**b**) Proportion of errors as a function of the set mean (binned). *Error bars* represent standard errors. (**c**) RT variability (within-bin standard deviation) decreases linearly with the absolute distance of the set mean from 5. (**d**) Response accuracy variability, measured as the Bernoulli variance $p(1-p)$ where p is the error rate, decreases with distance from 5

Heuristic alternatives replaced Δ_{ij} with either the extreme-deviation predictor $\text{maxDev}_{ij} = \max_k |x_{ijk} - 5|$ (maxDev) or the majority-strength predictor $\text{majAbs}_{ij} = \left|\sum_k \text{sgn}(x_{ijk} - 5)\right|$ (majAbs). Thus, the heuristic model family was:

$$\text{logit}\, P(\text{error}_{ij} = 1) = \beta_0 + \beta_H\, H_{ij} + \boldsymbol{\beta}_{\text{sd}}^{\top}\mathbf{SD}_{ij} + u_{0j}, \quad (12a)$$

$$\log(\text{RT}_{ij}) = \gamma_0 + \gamma_H\, H_{ij} + \boldsymbol{\gamma}_{\text{sd}}^{\top}\mathbf{SD}_{ij} + v_{0j} + \varepsilon_{ij}, \quad (12b)$$

with $H_{ij} \in \{\text{maxDev}_{ij}, \text{majAbs}_{ij}\}$.

Model selection by AIC strongly favored the averaging predictor Δ over both heuristic predictors. For log RT on correct trials, AIC = 19867.38 (Delta) versus 20335.57 (majAbs) and 21266.10 (maxDev). For accuracy (logit mixed-effects model), AIC = 13908.55 (Delta) versus 14286.33 (majAbs) and 14677.43 (maxDev). A descriptive "cell" model (condition × variability × distance) provided a ceiling fit (RT: AIC = 19799.77; accuracy: AIC = 13859.69), but the Δ model captured most of this structure with far fewer degrees of freedom. Overall, these comparisons indicate that performance is best explained by the distance of the set mean from 5 rather than by an extreme-item or majority-strength heuristic.

Individual-level model comparisons

To examine whether the model-comparison results reflected a broadly shared pattern across observers, we repeated the critical analyses at the individual-participant level. First, we estimated the effect of mean distance separately for each participant. For accuracy, 39 of 40 participants showed the

predicted negative relation between Δ and error rate, indicating fewer errors as the set mean moved farther from the criterion. This consistency was reliable by an exact sign test, $p < .001$, and the median participant-level slope was −1.31. For response time, 37 of 40 participants showed the predicted negative relation between Δ and log RT, again reliable by an exact sign test, $p < .001$. The median participant-level RT slope was −0.17, and the mean slope differed reliably from zero, $t(39) = -9.11$, $p < .001$, 95% CI [−0.204, −0.130].

We then compared, separately for each participant, models based on mean distance, the most extreme digit, and the relative number of digits above versus below the criterion. For accuracy, the mean-distance model provided the lowest AIC for 31 of 40 participants; by comparison, the majority-based model was preferred for eight participants and the extreme-item model for one participant. More directly, the mean-distance model outperformed the extreme-item model for 39 of 40 participants and outperformed the majority-based model for 32 of 40 participants. The same pattern was obtained for response times: the mean-distance model provided the lowest AIC for 31 of 40 participants, whereas the majority-based model was preferred for eight participants and the extreme-item model for one participant. Again, the mean-distance model outperformed the extreme-item model for 39 of 40 participants and the majority-based model for 32 of 40 participants. Thus, although there was some individual variability in the relative contribution of heuristic cues, the dominant participant-level pattern favored mean-based processing.

Finally, we examined whether the mean-distance effect persisted at the individual level in the diagnostic regimes designed to undermine heuristic responding. In the majority-tie condition, where the number of digits above and below the criterion was balanced, 38 of 40 participants showed the predicted negative relation between Δ and error rate, and 34 of 40 showed the predicted negative relation between Δ and log RT. In the extreme-conflict condition, where the most extreme digit pointed to the incorrect side of the criterion, 37 of 40 participants showed the predicted negative relation between Δ and error rate, and 37 of 40 showed the predicted negative relation between Δ and log RT. All four sign tests were reliable, $ps < .001$. These findings indicate that the mean-distance effect was not only present at the group level, but was also broadly expressed across individual observers, including in the conditions in which the main heuristic alternatives were uninformative or misleading.

Extreme-choice behavior on conflict trials There were $N = 4912$ conflict_extreme trials in which the extreme item implied the opposite choice from the mean. A logistic mixed-effects model predicting choose_extreme from the experimental cell revealed robust cell-to-cell differences (AIC = 4490.9), with estimated probabilities of choosing the extreme side ranging from .057 to .339 across cells.

This pattern calls for a straightforward interpretation. When mean-based evidence was strong – large Δ and an agreeing majority – participants rarely followed the extreme item (as low as $p = .057$), indicating that the misleading cue was effectively suppressed. Conversely, when mean-based evidence was weak – $\Delta = 0.5$ with a tied majority – reliance on the extreme item rose markedly (up to $p = .339$, nearly one trial in three). Thus, the extreme value *did* influence decisions, but only conditionally: it acted as a fallback cue under uncertainty rather than a primary decision rule.

These findings are naturally accommodated by a *noisy averaging* account. When the aggregate signal Δ is strong, participants rely on the mean and correctly override the extreme item. When Δ is small, and majority information is uninformative, internal noise renders the mean estimate unreliable, and participants are drawn toward the most salient local cue – the extreme digit. Crucially, a pure extreme-value heuristic cannot explain this pattern, as it would predict uniform reliance on the extreme item regardless of Δ or majority structure. Instead, the context-dependence of extreme-item influence reinforces the conclusion that ensemble averaging is the primary strategy, with heuristic responding emerging only when mean-based evidence is insufficient to guide a confident response.

Discussion

In Experiment 2, we tested again the ability of participants to extract the set mean from an ensemble of numerals (Brezis et al., 2016, 2015, 2018; Levin, 1974; Rosenbaum, Glickman, & Usher, 2021). In this experiment, we could also test the averaging-versus-heuristic hypothesis directly. The results strongly supported the conclusion that participants represented the average of the set, and only rarely took advantage of the majority-count rule and/or extreme-deviation heuristics. The design of the experiment provided a strict prediction regarding the emergence of the mean-set distance effect. If indeed participants had relied on these heuristics, then this distance effect should have substantially decreased or even been abolished altogether. Moreover, different response patterns were predicted to emerge in conditions where majority information was uninformative (ties) or when the extreme item favored the wrong response (conflict trials). Instead, overall model comparisons showed that distance remained the best single predictor of performance, even when the pertinent heuristic cues were available and salient.

The analysis of conflict_extreme trials complements this conclusion. Participants did use the extreme-item heuristic,

but only in highly ambiguous cases, where the set mean was close to the criterion (e.g., tied-majority trials with dist = 0.5). This is consistent with the idea that participants represented the average of the set most of the time, but occasionally shifted to that heuristic when uncertainty was high. Thus, we can conclude that in cases where the evidence based on the mean is sparse, participants resort to salient local cues such as an extreme item (Tsetsos et al., 2012). Mostly, however, when mean-based evidence is strong or medium, participants are less likely to use heuristics.

Taken together, Experiments 1–2 strongly supported the conclusion that performance in our comparative judgment task was governed by the distance of the set mean from the fixed standard. We did find some evidence for heuristic usage (Gigerenzer & Gaissmaier, 2011), but only as a secondary, context-dependent strategy, not as the main mode of operation. Experiment 2 not only replicated the core set-mean distance effect and variability signatures recorded in Experiment 1, it also adduced new evidence, based on model comparison, against two prominent heuristic alternatives.

General discussion

The present study provides strong evidence that participants can represent the arithmetic mean of small sets of *symbolic* single-digit numbers. In two experiments, we showed that both accuracy and response times obeyed the ensemble-distance effect, which reflects a monotonic improvement in performance as the distance between the display's set mean and the fixed criterion increased. Because "averaging-like" performance can emerge in this task without recourse to genuine averaging representations, but simply due to the use of heuristics (e.g., reliance on extrema or majority sign relative to the criterion), we intentionally engineered our stimulus space to dissociate the two. Although such heuristics can, in principle, sustain above-chance responding, the data consistently supported a mean-based account. The heuristic cues emerged as secondary influences only under conditions of increased uncertainty. These findings can inform us on both mechanisms of ensemble averaging over symbols (Brezis et al., 2015), and theories of symbolic magnitude representation (Dehaene, 2011; Nieder et al., 2002; Verguts & Fias, 2004). These findings can also speak to broader questions about how decision systems combine multiple cues (Fitousi & Noyman, 2024; Tsetsos et al., 2012).

Possible cognitive mechanism

The present results should not be taken to imply that observers explicitly calculated the arithmetic mean on each trial. Participants were instructed to respond without deliberate calculation, and the response times indicate that performance was too rapid to depend on a slow serial arithmetic procedure. Rather, we interpret the results as evidence for a rapid approximate integration process. This interpretation is consistent with models of ensemble perception in which observers extract summary statistics from sets of visual objects without representing each item with full precision (Alvarez, 2011; Alvarez & Oliva, 2009; Chong & Treisman, 2003; Whitney & Yamanashi Leib, 2018). In such models, the visual system can represent the central tendency of a set, such as its average size, orientation, spatial position, or other perceptual dimensions, even when memory for individual elements is limited (Dakin & Watt, 1997; Haberman, Harp, & Whitney, 2009; Parkes, Lund, Angelucci, Solomon, & Morgan, 2001).

The present task differs from classic ensemble-perception paradigms in that the relevant dimension was symbolic numerical magnitude rather than a continuous perceptual feature. However, classic models of numerical cognition provide a natural bridge between symbolic input and analog magnitude integration. The triple-code model, for example, assumes that Arabic numerals can be transformed into an analog magnitude code, especially when the task requires approximate quantity judgments (Dehaene, 1992). Similarly, neural-network models of number representation propose that both symbolic and nonsymbolic numerical inputs can activate overlapping magnitude representations that exhibit distance-dependent generalization (Verguts & Fias, 2004). Thus, although the stimuli in the present experiments were symbolic digits, their processing may have rapidly engaged analog magnitude representations.

On this account, each digit activates an approximate symbolic-magnitude representation, and these activated magnitudes are pooled into a summary signal that is compared with the reference value. Because the displays contained only four simultaneously presented single-digit numbers, such a process could operate without explicit summation and division. The distance between the set mean and the criterion then indexes the strength of the aggregate evidence: when the mean is far from 5, the pooled signal strongly favors one response, producing faster and more accurate judgments; when the mean is close to 5, the evidence is weaker, producing slower and less accurate judgments.

This interpretation also clarifies what is meant by "mean-based" performance in the present study. The arithmetic mean provides the formal task-relevant quantity and the best behavioral predictor, but the underlying psychological process need not compute the exact mean in an explicit mathematical sense. Instead, participants may approximate the mean through parallel symbolic-to-magnitude transla-

tion, noisy pooling, or evidence accumulation over the four values. This account remains compatible with the broader debate over whether apparent ensemble averaging reflects a dedicated averaging mechanism or can be approximated by focused-attention and subsampling strategies (Myczek & Simons, 2008). The present contribution to this debate is that the stimulus design allowed us to test mean-based predictions against specific shortcut accounts, including majority- and extreme-item strategies.

Future studies could test this account more directly by combining the present diagnostic stimulus design with post-task strategy reports, measures of numeracy or mathematical expertise, and process-tracing methods. Although self-reports are necessarily imperfect, they could provide useful complementary information about whether participants experience the task as intuitive averaging, approximate comparison, partial calculation, or some mixture of these strategies.

Relations to averaging

The present paradigm differs in several important respects from previous work on symbolic averaging. The majority of previous studies (Brezis et al., 2016, 2015, 2018; Rosenbaum, Glickman, & Usher, 2021) used sequential streams of two-digit numbers and often asked for a direct magnitude estimation of the average. In contrast, our task presented participants with a static 2×2 array of four *single-digit* numbers and asked them to judge whether the set mean was smaller or larger than a fixed criterion (5). We assessed performance via response times (RT) and accuracy. The novel combination of tailored stimulus space and a criterion-based response mode provided a cleaner and more diagnostic test of mean-based integration for several reasons.

First, our exclusive reliance on single-digit stimuli adds theoretical clarity that does not exist in previous studies that employed mostly two-digit numbers. Note that two-digit numbers are more complex than single-digit numbers because they require syntactic composition. The consensus among numerical cognition researchers is that single-digit numbers are processed rapidly in an analog/holistic manner (Dehaene, 2011). However, the processing of two-digit numbers is still debated. It is not clear whether they are processed holistically (Dehaene et al., 1990), analytically via decade–unit decomposition (Nuerk et al., 2001), or in a hybrid fashion (Fitousi & Algom, 2020). Consequently, paradigms that use two-digit numbers add uncertainty about the *within items* integration process, which can bias inferences about *across items* integration. Our exclusive reliance on single digits allows a more targeted assessment of cross-item integration.

Second, the static display reduces memory and order confounds inherent to sequential paradigms. Sequential streams recruit working memory because observers must encode, maintain, and update a running representation of values over time, and they invite temporal weighting. Indeed, recency effects have been documented in sequential numerical averaging (Brezis et al., 2015; Tsetsos et al., 2012). In our task, all items remained visible until response. This feature eliminates temporal order by design and reduces the contribution of order-dependent weighting and memory dynamics. The task is a cleaner assay of integration *per se*, less confounded by individual differences in memory capacity or by decay/interference processes that can be mistaken for averaging effects. Notably, Brezis et al. (2015) proposed a dual-route account of symbolic averaging. The model predicts that minimizing the sequential encoding procedure should induce less analytic computational routines in favor of a more intuitive component. Our task does exactly this, and as such assesses the intuitive component under conditions closer to parallel ensemble perception.

Third, the comparison with a fixed standard that we use yields a more diagnostic response than direct estimation tasks (e.g., sliders). Direct estimation tasks can conflate the internal representation of the mean with noise from motor planning, scale use, and response calibration. In contrast, the comparative judgment task assesses mean representation against a fixed standard. It measures its precision through RT and error rate, which generates a clear prediction with respect to mean-distance evidence, $\Delta = |\bar{x} - 5|$. Thus, our task, although using indirect measures, directly tests whether behavior is governed by the aggregate value of the set.

Fourth, the current design provides strong evidence against the possibility that the observed averaging-like patterns were generated by heuristics. This evidence cannot be readily adduced with previous paradigms. For example, in two-set comparisons with streams of equal set sizes (Tsetsos et al., 2012), participants can technically compare the sums rather than the means. In longer streams, participants can approximate the mean by additional heuristics such as subsampling, overweighting salient extremes, anchoring to landmarks, or relying on especially diagnostic samples. Thus, existing paradigms do show that participants can summarize symbolic sets, but they are less suited to adjudicate whether participants integrate all items into a mean representation. Our task affords a more stringent inference because it uses a fixed standard and therefore a continuous diagnostic predictor (mean-distance evidence), and it pits the set mean directly against heuristics. In Experiment 2, we purposefully built the stimulus space to make heuristic cues uninformative (majority ties) or misleading (extreme-item conflict), while matching mean distance and dispersion across conditions. The mean-distance effect was still persistent under these conditions, and model comparison adduced additional evidence that participants deployed mean-based evidence rather than heuristics.

A fifth consideration is that in a fixed set-size mean-comparison task, explicit summation is logically possible because $\bar{x} > 5$ is equivalent to $\sum_i x_i > 20$ for four items. However, this logical possibility does not by itself constitute a distinct cognitive account. The important issue is whether participants relied on deliberate, algorithmic arithmetic versus approximate magnitude integration (Brezis et al., 2015) – a concern that is especially relevant here because the display remained visible until response and could in principle support serial inspection or exact computation. Importantly, our conclusions do not hinge on assuming a particular encoding regime (parallel "gist" vs. serial sampling), but on identifying the decision variable that best accounts for trial-by-trial behavior. Participants were instructed to avoid calculating and to respond based on their impression of the mean; RTs and error rates exhibited a smooth, monotonic distance function with Δ, which also points to the absence of speed–accuracy trade-off; and mean-based predictors remained dominant even when salient shortcut cues were neutralized or made misleading. Accordingly, we treat explicit summation as an upper-bound strategy that is possible in principle but unlikely to explain the dominant pattern of results, while allowing that more deliberate computations may occasionally be recruited under uncertainty.

Finally, many earlier studies used direct estimation procedures (e.g., slider/ruler responses), which can be unreliable and susceptible to contextual and strategic biases (Poulton, 1979). The present indirect, comparative judgment – anchored by RT and accuracy – helps circumvent these limitations and provides a more diagnostic test of what participants compute.

Limitation We do not claim to have exhaustively ruled out every conceivable strategy; rather, the contribution of the present study is to separate mean-based integration from a theoretically motivated set of prominent heuristic alternatives and to show that a mean-based signal is required to explain performance across diagnostic conditions.

Implications for symbolic-to-analog mapping and numerical cognition

Many numerical cognition researchers (Dehaene et al., 1990; Fitousi, & Algom, 2020; Moyer, & Landauer, 1967; Piazza, Pinel, Le Bihan, & Dehaene, 2007; Verguts, & Fias, 2004) believe that Arabic numerals are mapped onto an analog magnitude representation. Our finding of a robust mean-set distance effect is consistent with this view. It shows that when people view a set of numerals, they likely transform each one of them into an analog quantity. In turn, these quantities are averaged, and the average quantity, which is also noisy, is compared with the representation of the standard. The emergence of a monotonic distance effect supports such a process. Our finding that the set's dispersion also affects performance is telling, because it supports the idea that the mapping becomes noisier when ensemble variability is heterogeneous.

Our findings support the notion of compositionality of numerical magnitudes, which has been important in number processing. This interpretation also relates to computational models of multidigit and multisymbol number comparison. Earlier work on two-digit number comparison showed that responses are affected not only by the overall numerical value of a two-digit number, but also by the compatibility between its component digits (Nuerk et al., 2001). More recent accounts have generalized this idea within broader models of multisymbol number comparison. In particular, Huber, Nuerk, Willmes, and Moeller (2016) proposed a general framework in which multisymbol numbers are processed through componential representations, with different symbolic components contributing to the comparison process according to their task relevance and salience. Although the present stimuli were not multidigit numbers in the place-value sense, the task similarly required observers to transform multiple symbolic components into magnitude-related evidence. The analogy is therefore not that a four-digit array is processed as a multidigit number, but that both tasks involve the coordination of several symbolic magnitude signals within a single comparison decision.

Finally, another implication of our work concerns mathematical competence (Siegler et al., 2012). Mathematical skills extend beyond simple operations on single digits, but even such simple abilities turn out to be a prerequisite for future success in more complex operations, including addition, multiplication, and multi-item integration (Siegler, Fazio, Bailey, & Zhou, 2013; Vamvakoussi, & Vosniadou, 2004), and for later mathematical achievement more broadly (Bailey, Hoard, Nugent, & Geary, 2012; Siegler et al., 2012). The ability to combine multiple symbolic magnitudes stands at the heart of many more complex operations, such as fractions (Fitousi & Noyman, 2024). These findings add to the numerical-cognition literature by showing that people can integrate small sets of Arabic digits and compute the average quickly and relatively efficiently.

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Data Availability Data and stimuli are available at https://osf.io/xnq92.

Code Availability Analysis code is available at https://osf.io/xnq92.

Declarations

Conflicts of Interest I have no conflicts of interest. I have no competing interests.

Ethical Approval The experiment received ethical approval.

Consent to Participate Participants gave their written consent to participate.

Consent for Publication This is not relevant for the present study.

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